Taking ACUE seriously as an object: a one-page brief

For Terence Tao, from Bill (Qingyun) Sun's programme with GPT-6 Astra and Fable. November 2026. Full statements and proofs: `NEW_RESULTS.pdf` (12 pages) in the same folder.

The starting point is your 2019 post. ACUE was built to show that everything currently provable about the zeros (Montgomery's form factor on $|\alpha| < 1$, hence every band-limited correlation of every order) is matched by a half-lattice model, so those inputs alone can never deliver GUE. We took the construction as an object in its own right and asked two questions. How large is the space of such impostors? And is there a natural functional of a configuration, not a band-limited correlation, that sees the lattice anyway? Everything below is proved or exactly computed, nothing conditional is presented as proved, and no famous conjecture is claimed.

1. ACUE is one point of an explicit family of impostors (the construction). Fix N. The support is the set of N-subsets C of the 2N-th roots of unity, with $\mu_{ACUE}(C)$ proportional to $|\Delta(C)|^2$. Call a statistic protected if it is a balanced polynomial of degree $\le N$ in the coefficients of the characteristic polynomial (equivalently a product of traces of total weight $\le N$); your theorem is that ACUE and CUE agree on all of them. Let $X(C) = \Sigma_{c\in C}\, c \pmod N$, the centre of mass on the lattice; up to a fixed sign, $\det U_C = e^{2\pi i X/N}$. Under ACUE, X is exactly uniform on $\mathbb{Z}/N$. For any $g: \mathbb{Z}/N \to [0, \infty)$ put

$$q_g(C) = \mu_{ACUE}(C) \cdot g(X(C)).$$

Theorem. q_g agrees with CUE on every protected statistic if and only if $E_{ACUE}\, g(X) = 1$ and $\hat{g}(\pm 1) = 0$. Proof idea: a protected moment moves a total "hop budget" of 2N steps around the lattice, while coupling to a centre-of-mass frequency $k \in \{2, \ldots, N-2\}$ costs at least 2N+2 steps, so the coupling vanishes. This is an $(N-3)$-parameter family of genuine probability measures on the same support as ACUE, every one of them indistinguishable from CUE by anything currently provable about zeros. The chiral members $g = 1 + \varepsilon \sin(2\pi k X/N)$, $2 \le k \le N-2$, $k \ne N/2$, $|\varepsilon| < 1$, sit at total-variation distance about 0.3 from ACUE yet give exactly the same law as ACUE to every reflection-invariant statistic. The full space of moment-matching signed measures on the support has dimension 0, 0, 2, 10, 80, 403 (or 399; unsettled), 1804 for $N = 3, \ldots, 9$, with positive members from $N = 5$ on. Why this matters to you: the functional equation's reflection symmetry kills exactly the chiral half of this fibre and provably nothing else. Any statistic proposed to refute AH has to be tested against the whole reflection-symmetric fibre, not against ACUE alone.

2. A first-passage time that sees the lattice. Run the finite heat flow on the characteristic polynomial, $P_s(z) = \Sigma_j\, a_j\, e^{s\cdot j(N-j)}\, z^j$, which is the finite analogue of the de Bruijn–Newman flow in the direction that destroys real-rootedness, and let the depth D be the first time two zeros collide (a finite analogue of $-\Lambda$). Theorems: $D \ge \delta_{min}^2/8$ for every configuration; $N^2 D \ge \pi^2/8$ for every non-clock ACUE configuration, while the alternating clock has $D = \infty$; for Haar CUE, $N^{8/3} D$ converges in law to $G^2/8$ with $P(G > x) = \exp(-x^3/72\pi)$. So CUE collapses at scale $N^{-8/3}$ and ACUE at scale N^{-2}: a clean separation. Along the flow the coefficients rescale deterministically, so every protected statistic is frozen (Astra showed the forward Coulomb flow preserves the protected algebra too). The depth is therefore invisible to the whole protected algebra, statically and dynamically, yet it is an honest functional of the point configuration. Exact constants: the simplest lattice defect (one extra point at a clock midpoint) has $\lim N^2 D = 2$ exactly, with full expansion $2 - 4/(3N^2) + \ldots$; the natural conjecture $N^2 D < 2$ on the whole AH support fails first at $N = 12$, at the gap pattern $[1^7, 2, 2, 9, 2, 2]$ in units of π/N, where $N^2 D = 2.0000177$; the supremum appears to be a domain-wall constant $C^* = 2.11466\ldots$, the first double-zero time of the heat flow of $\cos(u/2)/\Gamma(u/2\pi)$ (a half-line run of consecutive points joined to a half-line clock), matched to seven digits by lattice extrapolation. Why this matters to you: it is the first statistic we know of that is blind to every band-limited correlation but not to the lattice, and it lives in the same heat-flow world as your work with Rodgers.

3. A conservation law that explains why the impostors are hard-core. For any integer-marked configuration on $\mathbb{Z}/2N$ with N points that matches the CUE ramp $E|p_k|^2 = k$ for $1 \le k \le N-1$, all multiplicity is stored in the single unconstrained Fourier row: $E\, \Sigma_x\, m_x(m_x - 1) = (E|p_N|^2 - N)/2N$, and the fraction of simply-occupied sites is at least $\frac{1}{2} + 2/\pi^2 = 0.7026$ in the limit, with equality exactly on {0, 1, 2}-hard-core configurations. Collisions can only live at the Nyquist frequency. On the line there is no such frequency, which is why the finite lattice has a rigidity that the continuum pair-correlation problem lacks.

4. What this says about the goal, honestly. Astra reduced an AH refutation to one inequality: under RH, $\liminf_T W_T \ge 1/16$ for an explicit two-scale mean square of ζ'/ζ (AH gives $W = 0.06239$, GUE gives 0.08227). We then proved a barrier: $W_T = \int K(\alpha)\, F_T(\alpha)\, d\alpha + o(1)$ with an explicit kernel that changes sign at $\alpha_0 = \log(2 \cosh 1) = 1.127$, and every input the programme can prove (Montgomery on the band, positivity, the Goldston–Gonek–Özlük–Snyder floor under GRH) is satisfied by F_{AH}. So the separation above, and every band-limited argument, cannot close the gap. The last mile is exactly one out-of-band fact: a weighted average of F over $(1.13, \infty)$ must be at most 1.034 (AH gives 1.051, GUE gives 1.000). In gap language: AH puts almost every normalised gap at a half-integer, the proven small-gap bound under RH is $\mu \le 0.515396$ (Preobrazhenskiĭ), and the whole question sits in the three percent between 0.515 and 0.5. Nothing here touches RH itself; both ensembles live on the critical line.

Five questions we hope you will ask

Each is a direction where we think concrete progress is possible with the tools we already have.

1. "What is the fibre, really?" Is every moment-matching measure on the lattice a tilt of ACUE by a function of det U together with the reflection class, and what is the dimension of the fibre as a function of N? We can settle N = 8 exactly and attack a dimension formula by decomposing the protected algebra under the dihedral group acting on N-subsets of $\mathbb{Z}/2N$. A formula would say how much of your construction is forced by the moment constraints and how much is free, and it would tell everyone exactly which impostors a future argument has to defeat.

2. "Is the depth the right functional, or just the first one?" The depth separates because the lattice has minimum gap π/N while CUE has gaps down to $N^{-4/3}$. Is there a characterisation of all functionals that are continuous in the configuration, frozen on the protected algebra under both flows, and still separate the two ensembles? We conjecture every such functional factors through small-gap statistics, and we can test this, and probably prove it, on the reflection-symmetric fibre at small N. If true it says precisely that refuting AH means producing gaps below one half, with no subtler route available.

3. "Prove the domain-wall constant." The supremum of N^2D over the AH support should be $C^* = 2.11466\ldots$, approached from below along run|clock|hole configurations. The 3-block case (exactly 2) is proved by a local Gaussian-convolution limit plus the comparison theorem that confines every other gap. The missing lemma is a comparison inequality for the first-colliding pair: the wall protects its neighbours least. A sharp constant of this kind would be a new exact invariant of your ensemble, and it is the kind of problem an afternoon with you could settle or redirect.

4. "Can Rodgers–Tao be run on the half-lattice?" Your proof that $\Lambda \geq 0$ shows that $\Lambda < 0$ would force the zeros into local approximate arithmetic progressions, which is incompatible with Montgomery's pair-correlation information. The half-lattice is exactly the finest progression that information permits. Is there a quantitative window version, an effective Λ for a block of N consecutive zeros, whose first collision time under AH is at scale N^{-2} while under GUE it is at scale $N^{-8/3}$? Any explicit-formula handle on the first collision of a window, however weak, would be the first non-band-limited input anyone has, and we can compute both predictions exactly to test any proposed dictionary.

5. "Which prime-side statement is exactly the missing input?" By the Goldston–Montgomery equivalence, the form factor beyond the band is the variance of primes in short intervals. AH beats GUE in the barrier functional only through its Bragg atoms at even integers, so AH predicts a specific excess variance at specific interval lengths. We can compute exactly what AH predicts for the short-interval prime variance and at which lengths it departs from the GUE prediction, turning "refute AH" into one explicit statement about primes. The question we cannot answer alone is whether any known technology gives upper bounds for F averaged beyond the band, and that is the question we would most like to hear your view on.